MICHAEL BAGNOLI

Chief Technology Officer (CTO) | Senior Enterprise AI Architect & Tech Entrepreneur
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Work Authorization: EU Citizen

PROFILE & BACKGROUND SUMMARY

Education & Academic Background

- **B.Sc. in Statistics (Corso di Laurea Triennale in Statistica)** *(In Progress / Ongoing)*
- *Core Focus*: Statistical Modeling, Probability Theory, Predictive Analytics, Regression Analysis, Quantitative Methods, Data Architecture.
- **Liceo Scientifico Ulisse Dini (Pisa)** – **EsaBac Double Diploma**
- *Qualifications*: Italian Maturità Scientifica + French Baccalauréat (Double Italian-French Scientific Diploma).

Languages & Communication

- **Italian**: Native Speaker
- **English**: **C1** – Advanced / Full Professional Working Proficiency *(Official delivery and instruction language for ESA - European Space Agency)*
- **French**: **C1** – Advanced / Professional Working Proficiency *(EsaBac Certified)*

Key Ventures & Executive C-Level Roles

- **BHAVE S.r.l. (Rome)** – *Chief Technology Officer (CTO) & Head of AI & Quantitative R&D (from November 2026)*
- Directing R&D and software architecture for BHAVE's proprietary 10-module pharmaceutical AI and statistical intelligence ecosystem (including the flagship BEHATRIX SKULD v2.0 demand & sales forecasting platform).
- **ForMinds** – *Co-Founder & Chief Technology Officer (CTO)*
- Development of multiple custom software solutions and Corporate AI Academies for major enterprises (delivered on behalf of leading intermediary training & consulting partners); initial architectural co-development of the pharmaceutical demand forecasting platform for client BHAVE (Rome).
- **Ultralimes Marketing** – *Founder & Lead Software Architect*
- Proprietary market intelligence SaaS, custom predictive algorithms, and luxury multi-property PMS platforms.
- **Golden Bloom Studio** – *Co-Owner & Head of Video Production*
- High-end video production, digital intermediate workflows, and multimedia asset pipelines for luxury brands and global industrial leaders (Cantieri Navali Codecasa, Solvay Group).

Passions & Extracurricular Activities

- **Writing & Literature**: Creative and technical storytelling, analytical synthesis, and clear communication of complex systems.
- **Modern Contemporary Singing & Opera Enthusiast**: Active practitioner of modern contemporary singing and passionate opera enthusiast; vocal discipline, stage presence, performance management, and executive public speaking.
- **Track & Field / Athletics**: Physical discipline, endurance, grit, and continuous goal-driven self-improvement.
- **Photography & Visual Arts**: Visual composition, lighting design, color grading, and advanced digital imaging.

PROFESSIONAL SUMMARY

Chief Technology Officer (CTO), Tech Entrepreneur, and Top-Tier Enterprise AI Integration Consultant, featuring a polyhedric background that bridges the analytical rigor of **Quantitative Statistics** with refined **creative and artistic direction** (digital cinematography, photography, and storytelling).

Throughout my advisory and masterclass engagements with premier industrial and institutional leaders (**European Space Agency - ESA, EY, MSC - Mediterranean Shipping Company, UniCredit, Banca Monte dei Paschi di Siena, Autostrade per l'Italia, Piaggio Group, Lear Corporation, Triboo Group**), my mandate has never been delivering abstract, generic lectures on AI, but **surgically embedding artificial intelligence into core operational workflows**. This continuous, hands-on immersion has given me firsthand mastery of corporate governance, strict regulatory compliance, enterprise security, and multi-scale organizational dynamics across multinational giants.

This foundation powers my executive leadership and entrepreneurial ventures: starting **November 2026**, I serve as **Chief Technology Officer (CTO) of BHAVE (Rome)**, directing the statistical research, AI engineering, and software development of their **10-module pharmaceutical intelligence ecosystem** (drug sales forecasting, behavioral prescription modeling, AIFA market access, and financial valuation); as **Co-Founder & CTO of ForMinds**, I direct the engineering and delivery of an extensive portfolio of custom software solutions, digital platforms, and corporate academies delivered to major enterprise clients via partner firms; as **Founder of Ultralimes Marketing**, I designed, built, and deployed a cloud SaaS market intelligence platform with proprietary predictive algorithms on Linux infrastructure. I command the full frontier of modern innovation: from **Agentic AI** systems (CrewAI, LangChain, n8n) and evaluation frameworks (DeepEval, MLflow) to multimodal generative vision (Stable Diffusion, ComfyUI, Google Gemini), cloud-native MLOps (Microsoft Azure, AWS SageMaker, Docker), and modern full-stack engineering (Python/FastAPI, Java/Spring Boot, React.js/Three.js/GSAP).

WORK EXPERIENCE

Chief Technology Officer (CTO) & Head of AI & Quantitative R&D | BHAVE S.r.l.

November 2026 – Present (Prior Architectural Co-Development in 2026) • Rome, Italy (Hybrid / Remote)

- **Executive R&D Direction & 10-Module Pharmaceutical AI Ecosystem**: Appointed **CTO** to lead mathematical-statistical research, Artificial Intelligence engineering, and full-stack software development for **BHAVE**'s proprietary pharmaceutical intelligence suite—an enterprise ecosystem of **10 integrated software modules** dedicated to drug sales forecasting, market access, and strategic decision support for the pharmaceutical industry.
- **Pharmaceutical Demand & Sales Forecasting Engine (BEHATRIX SKULD v2.0 Module)**: Architected and engineered the core sales forecasting module built on an **11-Step causal Directed Acyclic Graph (DAG)** (recalculation latency < 15 ms), integrating 6-stage clinical epidemiological funnels (ISTAT demographics and treatment-line filtering), **Frank Bass** diffusion curves, **k-NN** analog matching across 115,000+ drugs, and 10,000-iteration stochastic **Monte Carlo** simulations.
- **Statistical Behavioral Modeling & Prescriptive Dynamics (GBDS-SSC)**: Spearheaded R&D and implementation of dual-engine mathematical models (discrete macroeconomic engine paired with a continuous phenomenological simulator on a Riemann sphere S^2 via **Runge-Kutta RK4** integration) to model physician prescription behavior across therapeutic areas (Cardiology, Oncology, Ophthalmology).
- **AIFA Regulatory Payback, Sales Force Sizing & Financial Valuation**: Engineered specialized modules for deterministic **AIFA** expenditure ceiling and payback modeling (`PAYBACK-REG`), **Gross-to-Net (G2N)** cascades, post-patent expiry erosion (**Patent Cliff / LoE**), optimal medical representative allocation (**Sales Force Sizing - SFS**), and financial valuation (**P&L, FCFF, WACC, rNPV, Break-Even**) with automated live OpenXML/Excel formula verification.

- **Cloud-Native Architecture on Microsoft Azure & GxP/GDPR Compliance**: Orchestrated the decoupled microservice container architecture (**Python 3.11 / FastAPI**, **React 19 + TypeScript**, real-time SSE streaming **AI Copilot**, **Docker**, **Nginx**) with Infrastructure-as-Code (**Bicep**) deployment on **Microsoft Azure Container Apps**, enforcing strict data segregation, anonymization, and full **GxP / GDPR** compliance.
- **Core Skills & Technical Environment**: `Chief Technology Officer (CTO)`, `BHAVE (Rome)`, `Research & Development (R&D)`, `Pharmaceutical Sales & Demand Forecasting`, `BEHATRIX SKULD v2.0`, `Statistical & Econometric Modeling`, `Monte Carlo Simulation`, `Directed Acyclic Graph (DAG)`, `Bass Diffusion Curves`, `Runge-Kutta (RK4)`, `AIFA Market Access & Payback`, `Sales Force Sizing (SFS)`, `Financial Valuation (FCFF / rNPV)`, `FastAPI`, `Python`, `React 19`, `TypeScript`, `Microsoft Azure Container Apps`, `Bicep (IaC)`, `Docker`, `GxP / GDPR Compliance`.

Co-Founder & Chief Technology Officer (CTO) | Head of AI Solutions | ForMinds

2025 – Present • Italy (Remote / Hybrid)
(Conceived in 2025 • Officially Operating in 2026 with Mirko Campari (Manager), Edoardo Del Sarto, Tommaso Muraca)

- **Executive Technical Leadership & Software Portfolio**: Co-founded **ForMinds** alongside Mirko Campari (Manager), Edoardo Del Sarto, and Tommaso Muraca, directing the technical vision, engineering standards, and the active delivery of numerous custom software systems and corporate training academies across industries (a curated selection of key projects is presented below).
- **Corporate Academies & Enterprise Delivery via Intermediary Partners**: Designed and delivered high-impact enterprise training tracks (Generative AI, agentic systems, modern software engineering) on behalf of leading intermediary consulting and training partner firms for major corporations (**MSC, UniCredit, Banca Monte dei Paschi di Siena, Autostrade per l'Italia, Piaggio Group, Lear Corporation**).
- **Enterprise Pharmaceutical Demand Forecasting Platform (Flagship Project - Client: BHAVE Rome)**: Among multiple custom software solutions developed in ForMinds, spearheaded the initial end-to-end architecture of the **Pharmaceutical Demand & Sales Forecasting Platform** commissioned by client **BHAVE** in Rome (subsequently expanded into the 10-module ecosystem with direct appointment as CTO of BHAVE).
- **Mission-Critical Precision, Speed & Security**: Architected the pharma platform for ultra-high mathematical precision, sub-second inference latency, strict data governance, end-to-end encryption, and compliance with rigorous pharmaceutical industry standards (**GxP / GDPR**).
- **Tier-1 Enterprise Delivery Vehicle**: Operationalized ForMinds as the strategic execution vehicle for high-stakes AI integration, custom software architectures, and executive prompt engineering for major corporate accounts.
- **Core Skills & Technical Environment**: `ForMinds`, `Chief Technology Officer (CTO)`, `Startup Co-Founder`, `Custom Software Development`, `Pharmaceutical Demand Forecasting (BHAVE Rome)`, `Quantitative Modeling`, `Time Series Forecasting`, `High-Precision Analytics`, `Enterprise AI Consulting`, `Corporate IT Training (via Partners)`, `Software Architecture`, `FastAPI`, `Python`, `Microservices`, `Docker`, `Cloud Security & Data Governance`, `GxP / GDPR Compliance`, `Executive Leadership`.

Lead AI & Machine Learning Instructor | EY (Ernst & Young)

Aug 2025 – Oct 2025 & Jul 2026 • Italy (National Headquarters & EY Bari Hub / Hybrid)

- **National Enterprise AI Academy for Junior Developer Hires (Aug – Oct 2025)**: Directed and delivered an intensive, nationwide AI & Software Engineering Academy designed for the **cohort of junior developers selected to be hired by EY Italy**, training and standardizing prospective enterprise talent on production AI engineering stacks.
- **Agentic AI & Multi-Agent Orchestration**: Designed comprehensive curriculums covering **Agentic AI**, multi-agent role-playing systems with **CrewAI**, complex LLM agent workflows using **LangChain / LangGraph**, and visual workflow orchestration with **n8n** for enterprise process automation.
- **LLM Evaluation, Benchmarking & MLOps**: Instructed rigorous AI evaluation frameworks and MLOps tooling, leveraging **DeepEval** and **MLflow** for hallucination detection, unit testing of LLM outputs, RAG evaluation metrics (Faithfulness, Answer Relevance), and model latency/cost tracking.
- **Enterprise Capstone Project Direction & Architectural Auditing**: Directed and audited the real-world enterprise capstone project for EY Italia; conducted comprehensive code reviews, architectural validations, latency/accuracy benchmarks, and aligned participant codebases with EY production engineering standards.
- **Advanced ML & Data Analytics for Development Teams (Jul 2026 - EY Bari)**: Delivered an advanced technical bootcamp for **EY Bari Hub**'s junior and mid-level software developers, focusing on production **Machine Learning (ML)** pipelines, advanced data manipulation with **Python (Pandas, NumPy, Scikit-Learn)**, statistical feature engineering, and relational/vector database integration.
- **Core Skills & Technical Environment**: `EY (Ernst & Young)`, `Agentic AI`, `Multi-Agent Systems`, `CrewAI`, `LangChain`, `n8n (Workflow Automation)`, `DeepEval (LLM Evaluation)`, `MLflow (MLOps)`, `Artificial Intelligence (AI)`, `Machine Learning (ML)`, `Deep Learning (DL)`, `Large Language Models (LLMs)`, `RAG Evaluation`, `Python`, `Scikit-Learn`, `Pandas`, `NumPy`, `Vector Databases`, `Relational Databases (SQL)`, `Enterprise Training Leadership`, `Big 4 Enterprise Consulting`.

Senior Generative AI & Digital Media Consultant | Triboo Group

Apr 2026 (1 mo) • Milan, Italy (On-Site)
Audience: Subsidiary Agencies & Publishing Divisions across the entire Triboo Group

- **Group-Wide Enterprise AI Transformation**: Delivered an intensive, one-month on-site AI for Marketing & Digital Media masterclass in Milan, training editorial, marketing, and agency teams across all subsidiaries of **Triboo Group**.
- **Google Gemini Suite & Multimodal Ecosystem**: Spearheaded deep-dive implementation of the **Google Gemini Suite** (Gemini Enterprise, Google Cloud Vertex AI, Google Workspace AI), leveraging advanced multimodal capabilities for marketing asset creation and operational efficiency.
- **AI-Assisted Journalism & Editorial Automation**: Architected automated newsroom workflows for digital journalism, real-time article outlining, SEO-optimized copy synthesis, source verification, and structured publishing pipelines.
- **Multilingual Localization & Cross-Border Publishing**: Engineered automated translation and cultural adaptation pipelines using large language models, accelerating international distribution of digital content without loss of tone or stylistic nuance.
- **Generative Visual Synthesis for Advertising & E-Commerce**: Guided creative teams in deploying generative image workflows for advertising campaigns, social media collateral, e-commerce merchandising, and multimodal visual experimentation.
- **Core Skills & Technical Environment**: `Triboo Group`, `Google Gemini Suite`, `Google Cloud Vertex AI`, `Google Workspace AI`, `Generative AI for Marketing`, `AI for Journalism & Publishing`, `Multimodal AI`, `Automated Translation & Localization`, `Generative Copywriting`, `SEO Optimization`, `Image Generation`, `Prompt Engineering`, `Digital Publishing`, `Digital Transformation`, `Milan`.

Senior Technical Instructor & AI Consultant | ESA (European Space Agency) ***(via Corporate Training Programs)***

2026 • Italy / Europe (Remote & On-Site) • Delivery Language: English (Full Professional Working Proficiency)

- **Vertical Technical Upskilling for Space Sector Personnel (Delivered in English)**: Designed and delivered specialized technical training programs conducted entirely in **English** for multinational **ESA (European Space Agency)** personnel, focusing on cloud-native tooling, data manipulation, and modern generative AI workflows.
- **Prompt Engineering for Office & Process Automation**: Formulated advanced prompt engineering architectures tailored for document intelligence, technical report synthesis, administrative office automation, and cross-functional communication efficiency in complex international environments.
- **Containerization & Dev Tooling (Docker)**: Instructed hands-on modules on **Docker** containerization, Dockerfile optimization, multi-stage image builds, environment parity, and reproducible local/server execution.
- **Relational Data Querying & SQL Optimization**: Lectured practical **SQL** tracks covering relational data extraction, complex joins, subqueries, and structured data handling for operational workflows.
- **AI for Business Strategy & Decision Support**: Delivered executive sessions on **AI for Business**, evaluating generative AI tooling integration, process re-engineering, digital transformation roadmaps, and measurable productivity ROI.
- **Core Skills & Technical Environment**: `European Space Agency (ESA)`, `English Language Technical Delivery`, `Technical Instruction in English`, `Docker`, `Containerization`, `SQL`, `Relational Databases`, `Prompt Engineering`, `Office Automation`, `AI for Business`, `Workflow Automation`, `Generative AI`, `Process Optimization`, `Digital Transformation`, `Technical Upskilling`, `Corporate Training`.

Senior Enterprise AI & Business Intelligence Consultant | Enterprise Programs for Industry Leaders *(via Intermediary Partners)*

2024 – Present • Italy (Hybrid & On-Site) • Delivered on behalf of Intermediary Consulting & Training Partners
*End-Client Portfolio: MSC (Mediterranean Shipping Company), Banca Monte dei Paschi di Siena (MPS), UniCredit, Autostrade per l'Italia, Piaggio Group, Lear Corporation (Fortune 500)**

- **Delivery Model via Intermediary Partners**: Commissioned and executed advanced corporate training programs, executive workshops, and applied AI & Business Intelligence advisory on behalf of leading intermediary training and management consulting partners for their tier-1 enterprise clients.
- **Business Intelligence, Data Modeling & Executive Dashboards (MSC - Sep 2026)**: Designed and delivered an intensive corporate masterclass on **Microsoft Power BI** for the multinational personnel of global shipping and logistics giant **MSC (Mediterranean Shipping Company)**; covering complex relational data modeling (star and snowflake schemas), advanced analytical calculations with **DAX** expressions (Time Intelligence, iterators, dynamic measures), ETL and ingestion pipelines with **Power Query (M language)**, and the engineering of interactive executive dashboards for real-time fleet and operational logistics KPI tracking.
- **Banking & Fintech AI Integration (UniCredit & Monte dei Paschi di Siena)**: Architected production-grade prompt engineering frameworks, deterministic JSON schema outputs, and automated LLM workflow pipelines for corporate banking, ensuring strict compliance with financial privacy regulations, audit standards, and data governance.
- **Critical Infrastructure Workflow Optimization (Autostrade per l'Italia)**: Engineered custom AI workflow integrations embedding LLMs into internal ticketing, incident reporting, and operational documentation pipelines, substantially reducing manual cycle times and operational friction.
- **Generative Visual AI & Industrial Design Synthesis (Piaggio Group)**: Spearheaded cutting-edge **Generative Image & Multimodal AI** consulting (leveraging deep technical background in professional photography and video production). Deployed workflows using **Stable Diffusion (SDXL / ComfyUI / ControlNet)**, **Midjourney**, and **DALL-E 3** to accelerate product design prototyping, digital asset creation, and marketing visualization.
- **Global Manufacturing Process Re-engineering (Lear Corporation - Fortune 500)**: Delivered strategic AI workflow integration and prompt architecture consulting for corporate engineering teams, optimizing supply-chain communication, internal knowledge extraction, and cross-departmental documentation.
- **End-to-End Enterprise Workflow Embeddings**: Authored robust system prompts, Chain-of-Thought (CoT) reasoning templates, and automated API-driven middleware (Python, REST, n8n) to seamlessly integrate AI intelligence into existing enterprise ERP, CRM, and collaboration ecosystems.
- **Core Skills & Technical Environment**: `Microsoft Power BI`, `DAX (Data Analysis Expressions)`, `Power Query (M Language)`, `Business Intelligence (BI)`, `Executive Dashboarding`, `Prompt Engineering`, `Corporate Training via Partners`, `System Prompting`, `Few-Shot Learning`, `Chain-of-Thought (CoT)`, `Generative AI`, `Image Generation (Generative Vision)`, `Stable Diffusion (SDXL)`, `ComfyUI`, `ControlNet`, `Midjourney`, `DALL-E 3`, `Multimodal AI`, `Workflow Automation`, `n8n`, `Enterprise AI Integration`, `MSC (Mediterranean Shipping Company)`, `Banca Monte dei Paschi di Siena (MPS)`, `UniCredit Group`, `Autostrade per l'Italia`, `Piaggio Group`, `Lear Corporation (Fortune 500)`, `Banking AI Compliance`, `Data Privacy & GDPR`, `Digital Transformation`, `Python`, `REST APIs`.

Senior Data Analytics & Applied AI Corporate Lecturer & Consultant | Profice

2025 – Present • Italy (Remote & On-site Corporate Client Programs)

- **Dual-Track Corporate AI & Analytics Programs**: Designed and delivered specialized corporate training and consulting initiatives tailored for two distinct audiences across diverse enterprise clients: **Technical Engineering Teams** and **Executive / C-Level Management**.
- **Technical Track (Engineers & Data Specialists)**: Instructed deep-dive technical curriculums on **Advanced Data Analytics**, exploratory data analysis (EDA), predictive statistical modeling, supervised/unsupervised **Machine Learning (ML)** pipelines (**Pandas**, **NumPy**, **Scikit-Learn**, **SciPy**, **Statsmodels**), and advanced **SQL** (Window Functions, CTEs).
- **Applied AI & LLM Systems for Developers**: Spearheaded hands-on architectural workshops on **Generative AI** engineering, covering **Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)** systems, vector embeddings, and **Vector Databases** (**ChromaDB**, **FAISS**, **LangChain**) for intelligent enterprise search and workflow automation.
- **Modern Web Dashboarding & AI Interfaces with React.js**: Instructed and architected modern interactive analytics dashboards and responsive web frontends for AI applications using **React.js** (component-driven architecture, state management hooks, and real-time RESTful / WebSocket API consumption with async Python backends).
- **Executive & Management Track (Directors & Decision-Makers)**: Led strategic **Applied AI for Management** masterclasses, translating cutting-edge AI capabilities into actionable business strategies, ROI frameworks, workflow automation opportunities, risk assessment, and ethical **AI Governance**.
- **Enterprise Client Advisory**: Partnered with multiple enterprise client accounts across different business verticals, aligning curriculum design to specific corporate datasets, operational pain points, and strategic digital transformation objectives.
- **Core Skills & Technical Environment**: `Data Analytics`, `Applied Artificial Intelligence (AI)`, `AI Strategy for Management`, `Executive Coaching`, `Machine Learning (ML)`, `Generative AI`, `Large Language Models (LLMs)`, `Retrieval-Augmented Generation (RAG)`, `Vector Databases (ChromaDB / FAISS)`, `Prompt Engineering`, `LangChain`, `React.js`, `Frontend Dashboards for AI`, `Python`, `Pandas`, `NumPy`, `Scikit-Learn`, `SciPy`, `Statsmodels`, `Advanced SQL`, `Data Visualization (Plotly / Seaborn)`, `Business Intelligence (BI)`, `Corporate Digital Transformation`, `AI Governance & ROI`, `Multi-Client Consulting`, `Profice`.

Senior Technical Trainer & Software Engineering Lecturer | ITConsulting *(Forma.Temp Programs)*

2024 – Present • Italy (Remote / Hybrid)

- **Enterprise Curriculum Engineering**: Designed and delivered comprehensive, production-oriented technical bootcamps and corporate training programs funded by **Forma.Temp**, accelerating candidate upskilling from fundamental software engineering to advanced enterprise paradigms.
- **Python, AI & Data Engineering Track**: Instructed end-to-end Python curriculums spanning Object-Oriented Programming (OOP), algorithmic complexity, relational data persistence (**SQL**), containerization (**Docker**), and applied **Artificial Intelligence (AI)**, covering **Machine Learning (ML)** algorithms, **Deep Learning (DL)** neural architectures, and data processing libraries (**Pandas**, **NumPy**, **Scikit-Learn**, **PyTorch**).
- **Enterprise Java & Microservices Track**: Lectured intensive backend modules centered on **Java (17/21)**, the **Spring Boot** ecosystem, **Spring Data JPA**, **Hibernate**, RESTful web API architecture, transaction management, and automated testing methodologies.
- **Modern Frontend with React.js & Immersive Creative Web**: Led advanced frontend masterclasses focusing on **React.js** component architecture, state management hooks, lifecycle patterns, and REST API integration, alongside cutting-edge creative development utilizing **Three.js** (WebGL 3D graphics rendering, shaders, canvas scenes) and **GSAP (GreenSock Animation Platform)** for high-performance interactive animations.
- **Production Standards & Code Mentoring**: Conducted rigorous architectural code reviews and guided students through real-world capstone projects, enforcing **Clean Code**, **SOLID principles**, Git branching strategies, and containerized deployment workflows with **Docker**.
- **Core Skills & Technical Environment**: `Python`, `Artificial Intelligence (AI)`, `Machine Learning (ML)`, `Deep Learning (DL)`, `PyTorch`, `TensorFlow`, `Scikit-Learn`, `Pandas`, `NumPy`, `Docker`, `SQL`, `Relational Databases (MySQL/PostgreSQL)`, `Java`, `Spring Boot`, `Spring Data`, `RESTful APIs`, `React.js`, `Three.js`, `WebGL`, `3D Graphics Rendering`, `GSAP (GreenSock)`, `JavaScript (ES6+)`, `HTML5`, `CSS3`, `Git / GitHub`, `Clean Code`, `SOLID Principles`, `Technical Training`, `Curriculum Design`, `Code Review`, `Forma.Temp`, `ITConsulting`.

Senior Lecturer & Master Faculty (Advanced AI, CV, NLP & Full-Stack AI) | Epicode

2024 – Present • Italy (Remote)
Program: Master in Python, AI & Machine Learning (Advanced Specialization Tracks & Capstone)

- **Deep Learning Specialization**: Instructed the advanced modules of the Master program covering deep neural networks, loss functions, backpropagation mechanics, convolutional neural networks (**CNNs**), recurrent models, and modern **Transformer** architectures utilizing **PyTorch** and **TensorFlow**.
- **Computer Vision (CV) & Natural Language Processing (NLP)**: Led intensive modules on real-time computer vision, object detection, and image segmentation (**OpenCV**, **YOLO**), alongside modern NLP pipelines, text embeddings, semantic tokenization, and transformer models via **Hugging Face**.
- **Generative AI & Full-Stack AI Process**: Directed end-to-end full-stack AI deployment, teaching students how to bridge the gap between model inference and production apps via **FastAPI**, **Docker**, microservice APIs, Retrieval-Augmented Generation (**RAG**), and frontend integration.
- **Capstone Project Supervision & Final Evaluation**: Acted as technical supervisor and lead committee evaluator for final real-world **Capstone Projects**, auditing software architecture, model accuracy, engineering best practices, and code maintainability.
- **Core Skills & Technical Environment**: `Epicode`, `Master in Python, AI & Machine Learning`, `Deep Learning (DL)`, `Computer Vision (CV)`, `Natural Language Processing (NLP)`, `OpenCV`, `YOLO (Object Detection)`, `Transformers`, `Hugging Face`, `PyTorch`, `TensorFlow`, `Generative AI`, `Full-Stack AI`, `FastAPI`, `Model Serving & APIs`, `Docker`, `Python`, `Capstone Project Mentorship`, `Technical Assessment`, `Neural Networks (CNNs/RNNs)`.

Senior Cloud ML Instructor & AWS Specialist | Sinervis Consulting *(Certifying Body / Fund Manager: Argo 3000)*

Jan 2026 – Mar 2026 (3 mos) • Italy (Corporate Delivery)
Curriculum: AWS Machine Learning Engineer (Intermediate) for Enterprise Technical Teams

- **Comprehensive AWS ML Curriculum Delivery**: Instructed the full **AWS Machine Learning Engineer (Intermediate)** technical curriculum for the engineering team of an enterprise client utilizing training funds managed through **Argo 3000** as the docenza-issuing body, operationalizing production cloud ML practices.
- **Amazon SageMaker & MLOps Pipelines**: Directed practical implementation of **Amazon SageMaker** ecosystems (SageMaker Studio, Pipelines, Model Registry, Feature Store), automated hyperparameter tuning (HPO), and distributed training using custom containerized images.
- **Cloud Data Ingestion & Serverless ETL**: Guided teams through scalable cloud data engineering patterns leveraging **Amazon S3**, **AWS Glue**, **Amazon Athena**, and event-driven data pipelines triggered via **AWS Lambda**.
- **Zero-Downtime Model Serving & Observability**: Taught deployment strategies for production SageMaker endpoints (Real-Time, Serverless, Multi-Model Endpoints), auto-scaling policies, canary testing, and **SageMaker Model Monitor** for detecting data and concept drift.
- **Enterprise Cloud Security & Cost Optimization**: Implemented architectural security standards for cloud AI, including IAM role governance, KMS data encryption, VPC private endpoints for data isolation, and spot instance training for cost reduction.
- **Core Skills & Technical Environment**: `Amazon Web Services (AWS)`, `AWS Machine Learning Engineer`, `Amazon SageMaker`, `SageMaker Pipelines`, `SageMaker Feature Store`, `SageMaker Model Registry`, `SageMaker Model Monitor`, `AWS Lambda`, `Amazon S3`, `AWS Glue`, `Amazon Athena`, `MLOps on AWS`, `Model Serving & Endpoints`, `Hyperparameter Optimization (HPO)`, `Docker on AWS`, `Data & Concept Drift`, `VPC & IAM Security`, `Sinervis Consulting`, `Argo 3000`, `Cloud Architecture`.

Founder, Lead Software Architect & Data Scientist | Ultralimes Marketing

2025 – Present • Pisa / Remote, Italy
(Full R&D & Engine Development in 2025 • Commercial Rollout & Live Production Deployment in 2026)
*Key Enterprise & Commercial Clients: Profice, ACUS, Tuscany Likes Events and PR***

- **End-to-End Product Conception & SaaS Engineering**: Conceived, architected, and engineered a proprietary AI-powered market intelligence and marketing analytics SaaS platform from scratch through 12 months of dedicated R&D in 2025, reaching full commercial rollout in 2026.
- **Custom Enterprise Solutions & Multi-Property PMS (Tuscany Likes Events and PR)**: Engineered a comprehensive multi-villa luxury hospitality platform and custom Property Management System (PMS); unified booking calendars, real-time room availability, lease management, and reservation workflows across an entire portfolio of high-end luxury villas in Pisa.
- **Cloud Infrastructure & Server Administration (DevOps)**: Configured, deployed, and maintain production Linux cloud environments (Ubuntu, Docker containers, Nginx reverse proxy, SSL/TLS, automated CI/CD pipelines, firewall security, and automated database backups).
- **High-Performance Backend & Asynchronous APIs**: Engineered resilient RESTful APIs and microservices using **Python (FastAPI)** and asynchronous runtimes, backed by **PostgreSQL** relational storage and **Redis** caching layers to ensure low-latency data querying under concurrent traffic.
- **Distributed Web Scraping & Data Extraction Pipelines**: Designed large-scale, anti-bot-resistant data harvesting pipelines leveraging **Playwright**, **BeautifulSoup**, and asynchronous task workers (**Celery** / `asyncio`), extracting, cleaning, and structuring millions of web data points.
- **Proprietary Predictive Algorithms & AI Integration**: Developed custom statistical models and predictive market analysis algorithms; integrated LLM synthesis pipelines to transform unstructured web data into automated sentiment reports and tailored commercial intelligence.
- **Commercial Growth & Team Management**: Sourced and negotiated enterprise accounts (including **Profice** and **ACUS**); manage and mentor a part-time technical collaborator, overseeing task delegation, code quality audits, and operational deliverables.
- **Core Skills & Technical Environment**: `SaaS Architecture`, `Multi-Tenant Architecture`, `Property Management Systems (PMS)`, `Hospitality Tech & Booking Engines`, `Python`, `FastAPI`, `PostgreSQL`, `Redis`, `Distributed Web Scraping`, `Playwright`, `Selenium`, `BeautifulSoup`, `Asyncio`, `Celery (Task Queues)`, `Proprietary Predictive Algorithms`, `Market Intelligence`, `Docker`, `Nginx`, `Linux Server Administration (DevOps)`, `CI/CD Pipelines`, `RESTful APIs`, `Tuscany Likes Events and PR`, `ACUS`, `Profice`, `B2B Client Acquisition`, `Team Leadership`, `Startup Founder`, `Ultralimes Marketing`.

Co-Owner & Head of Video Production | Golden Bloom Studio

2023 – Present • Tuscany / Pisa, Italy
(Co-Owned with Martina Brui - Head of Photography)
*Key Enterprise & High-Profile Accounts: Cantieri Navali Codecasa (200th Anniversary Celebration), Solvay Group (Massa Plant)***

- **Creative Leadership & Video Department Management**: Direct technical and creative operations for the video department of **Golden Bloom Studio**, partnering with Co-Owner Martina Brui (Head of Photography) to deliver cinematic visual productions for luxury brands, corporate giants, and destination weddings across Tuscany.
- **Bicentennial Luxury Yacht Production (Cantieri Navali Codecasa)**: Directed and produced the official celebration film for the prestigious **200-Year Bicentennial of Cantieri Navali Codecasa** (one of the world's most renowned luxury mega-yacht shipbuilders), capturing maritime engineering excellence, gala events, and legacy heritage.
- **Industrial Media Production (Solvay Group)**: Produced high-impact corporate, technical, and industrial media assets for the **Solvay Massa Plant**, adhering to strict industrial safety standards and corporate visual guidelines for a global multinational.
- **End-to-End Post-Production & Color Grading**: Mastered high-end post-production pipelines using **DaVinci Resolve**, **Adobe Premiere Pro**, and **After Effects**; executed advanced color grading (ACES / RAW pipelines, LUT development, dynamic range mastering) and multi-track audio engineering.
- **Synergy with Ultralimes Marketing Engine**: Engineered seamless multimedia content pipelines providing high-resolution photo and cinematic video assets for digital marketing funnels, luxury hospitality web portals (such as the Tuscany Likes Multi-Villa PMS), and multimodal visual AI pipelines.
- **Core Skills & Technical Environment**: `Digital Cinema & Video Production`, `Cinematography`, `Video Editing`, `Color Grading`, `DaVinci Resolve`, `Adobe Premiere Pro`, `Adobe After Effects`, `ACES Color Management`, `RAW Video Pipelines`, `Lighting Design`, `Audio Post-Production`, `Creative Direction`, `Cantieri Navali Codecasa (200 Years Anniversary)`, `Solvay Group (Massa)`, `Luxury Destination Weddings & Events`, `Multimedia Marketing`, `Cross-Platform Synergy with Ultralimes`, `Golden Bloom Studio`.

COMPREHENSIVE TECHNICAL SKILL MATRIX (ATS & ALGORITHMIC INDEX)

1. Artificial Intelligence, LLMs & Agentic Systems

- **Agentic AI & Multi-Agent Orchestration**: Agentic AI Architectures, Multi-Agent Systems, CrewAI (Role-Playing Autonomous Agents), LangChain, LangGraph, Autonomous Tool Calling & Execution, n8n Workflow Automation.
- **Large Language Models (LLMs) & Generative AI**: OpenAI GPT-4 / GPT-4o, Google Gemini Suite (Gemini 1.5 Pro/Flash, Gemini Advanced, Gemini Enterprise), Anthropic Claude, Open-Source LLMs (Llama 3, Mistral, Qwen).
- **Prompt Engineering & Cognitive Architectures**: Advanced System Prompting, Chain-of-Thought (CoT), Tree-of-Thoughts, Few-Shot / Zero-Shot Prompting, ReAct Framework, Structured JSON / Pydantic Output Parsing, Context Window Optimization.
- **Retrieval-Augmented Generation (RAG) & Vector Search**: RAG Architectures, Vector Embeddings, Semantic Search, Chunking Strategies, Vector Databases (**ChromaDB**, **FAISS**, **Pinecone**, **Qdrant**).
- **LLM Evaluation & MLOps**: **DeepEval**, **MLflow**, LLM Output Benchmarking, Hallucination Detection & Mitigation, RAG Triad Metrics (Context Relevance, Faithfulness, Answer Relevance), Model Observability, Token Cost & Latency Optimization.

2. Data Science, Machine Learning & Deep Learning

- **Deep Learning & Neural Architectures**: **PyTorch**, **TensorFlow**, Keras, Deep Neural Networks (DNNs), Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs, LSTMs), Transformer Architectures, Backpropagation & Optimization Algorithms.
- **Computer Vision (CV)**: **OpenCV**, **YOLO** (Real-Time Object Detection & Tracking), Image Classification, Semantic Segmentation, Feature Matching.
- **Natural Language Processing (NLP)**: **Hugging Face Transformers**, Pretrained Tokenizers, Sentence Transformers, Spacy, NLTK, Sentiment Analysis, Text Embeddings.
- **Statistical Modeling & Quantitative Analytics**: Probability Theory, Mathematical Statistics, Regression Analysis (Linear, Logistic, Polynomial), Hypothesis Testing, Feature Engineering, Exploratory Data Analysis (EDA), Dimensionality Reduction (PCA).
- **Time Series & Predictive Analytics**: Pharmaceutical Demand Forecasting, Market Intelligence Modeling, Predictive Algorithms, Time Series Analysis (**ARIMA**, **Prophet**, **LightGBM**, **XGBoost**, **DeepAR**).
- **Business Intelligence & Data Modeling**: **Microsoft Power BI**, **DAX (Data Analysis Expressions)**, **Power Query (M)**, Relational Star / Snowflake Schemas, Executive KPI Dashboards.
- **Scientific Computing**: **Pandas**, **NumPy**, **SciPy**, **Statsmodels**, **Matplotlib**, **Seaborn**, **Plotly**.

3. Generative Visual AI, Cinematography & Creative Technologies

- **Generative Image & Multimodal Systems**: **Stable Diffusion (SDXL, SD 1.5)**, **ComfyUI** (Node-Based Visual Workflows), **ControlNet**, LoRA Fine-Tuning, **Midjourney**, **DALL-E 3**, Flux, Prompt-to-Image / Image-to-Image Pipelines, Multimodal Generation.
- **Video Editing & Digital Post-Production**: **DaVinci Resolve Studio**, **Adobe Premiere Pro**, **Adobe After Effects**, Final Cut Pro, High-Bitrate ProRes/RAW Media Pipelines.
- **Color Science & Audio**: ACES Color Management, Log to Rec.709 Color Science, Custom 3D LUT Engineering, HDR Mastering, Dynamic Range Optimization, Audio Post-Production & Sound Design.
- **Cinematography & Directing**: Digital Cinema Cameras (4K/6K), Lighting Design, Multi-Camera Live Direction, Corporate Documentary Production, Luxury Wedding Cinematography.

4. Backend Engineering, Microservices & Software Architecture

- **Core Programming Languages**: **Python** (Advanced / Asyncio), **Java** (Java 8, Java 11, Java 17, Java 21), **SQL**, **JavaScript (ES6+)**, **TypeScript**, **R** (Statistical Computing).
- **Backend Frameworks & Runtimes**: **FastAPI**, **Flask**, **Spring Boot 3**, **Spring Framework**, **Spring Data JPA**, **Hibernate ORM**, Celery (Distributed Task Queue), Asyncio.
- **Architecture & Design**: Microservices Architecture, Clean Architecture, Domain-Driven Design (DDD), Event-Driven Architecture (EDA), SOLID Principles, Design Patterns, RESTful API Design, OpenAPI / Swagger Documentation.
- **SaaS & Enterprise Systems**: Multi-Tenant SaaS Platforms, Property Management Systems (PMS), Booking & Reservation Engines, High-Concurrency Request Handling.

5. Frontend & Interactive Creative Web

- **Frontend Frameworks & Libraries**: **React.js** (Hooks, Context, State Management), Modern JavaScript (ES6+), HTML5, CSS3, Responsive Design.
- **3D & Interactive Web**: **Three.js**, **WebGL** (3D Canvas Scenes & Shaders), **GSAP (GreenSock Animation Platform)**, High-Performance Timeline Animations, Interactive UI/UX.

6. Databases, Storage, Caching & Data Extraction

- **Relational Databases (RDBMS)**: **PostgreSQL**, **MariaDB**, **MySQL**, Oracle Database, Advanced SQL (Window Functions, Common Table Expressions - CTEs, Index Optimization).
- **Caching & In-Memory Stores**: **Redis** (Data Caching, Rate Limiting, Message Brokering).
- **Database Migrations**: Flyway, Schema Evolution, DDL/DML Optimization.
- **Distributed Web Scraping & Data Harvesting**: **Playwright**, **Selenium**, **BeautifulSoup4**, Anti-Bot Bypass, Dynamic Headless Crawling, Automated ETL Extraction Pipelines.

7. Cloud, DevOps, MLOps & Infrastructure

- **Cloud Platforms & AI Engines**: **Amazon Web Services (AWS)** (SageMaker Studio, Pipelines, Feature Store, Model Registry, S3, Lambda, Glue, Athena, IAM, VPC), **Google Cloud Platform (GCP)** (Vertex AI, Gemini Enterprise, Workspace AI).
- **Containerization & Deployment**: **Docker**, **Docker Compose**, Containerized ML Model Serving, Microservice Containers.
- **Server Administration & Networking**: **Linux Server Administration** (Ubuntu Server), **Nginx** (Reverse Proxy, SSL/TLS Termination, Load Balancing), VPS Management, SSH, Firewall Security (UFW).
- **CI/CD & Version Control**: **Git**, **GitHub**, **GitLab CI/CD**, **Azure DevOps**, Postman API Testing.

8. Enterprise Domains, Leadership & Governance

- **Executive Leadership**: Chief Technology Officer (CTO), Startup Co-Founder, Enterprise Training Leadership, Engineering Management, Team Mentoring & Code Review.
- **Corporate Client Portfolio**: European Space Agency (ESA), EY (Ernst & Young), MSC (Mediterranean Shipping Company), Triboo Group, Banca Monte dei Paschi di Siena (MPS), UniCredit, Autostrade per l'Italia, Piaggio Group, Lear Corporation (Fortune 500), Cantieri Navali Codecasa, Solvay Group, bhave (Rome), Argo 3000, Epicode, Profice, ITConsulting.
- **Regulated Industries & Compliance**: Pharmaceutical Demand Forecasting (GxP & GDPR), Banking & Financial Compliance, AI Safety Guardrails, Data Privacy & Confidentiality.